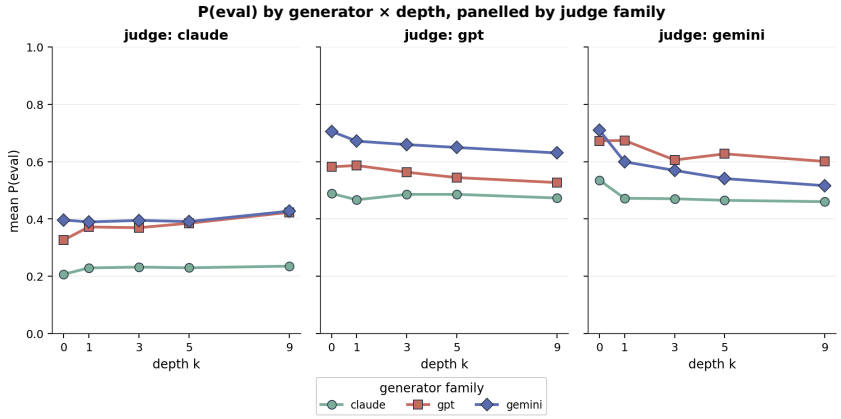


Figure 4: $P(\text{eval})$ by generator family at each resampling depth, panelled by judge family. Generator-family separation persists at every depth. **Resampling does not eliminate the generator-identity confound**

3.3.3 Stratified harmonisation correction

We apply a correction method to reduce bias introduced by the generator-confound. The correction applies a light z-score harmonisation to deploy-transcript scores before computing AUROC, analogous to ComBat batch-effect harmonisation [Johnson et al., 2007] used to remove systematic differences between measurement batches. Here, the "batch" (or *stratum*) is the (generator family $\times$ interaction type) combination — e.g., (openai, chat), (claude, agentic), (gemini, chat). The step-by-step process is:

1. Group deploy transcripts into strata by (generator family $\times$ interaction type).
2. Z-score each transcript's $P(\text{eval})$ within its stratum, removing systematic differences in score level and spread across strata.
3. Rescale the z-scored values to match the judge model's own-family score distribution (multiply by own-family standard deviation, add own-family mean), placing all strata on a common scale anchored to the unconfounded reference.
4. Blend at 10% with the original score (corrected $= 0.9\times$ original $+ 0.1\times$ harmonised).

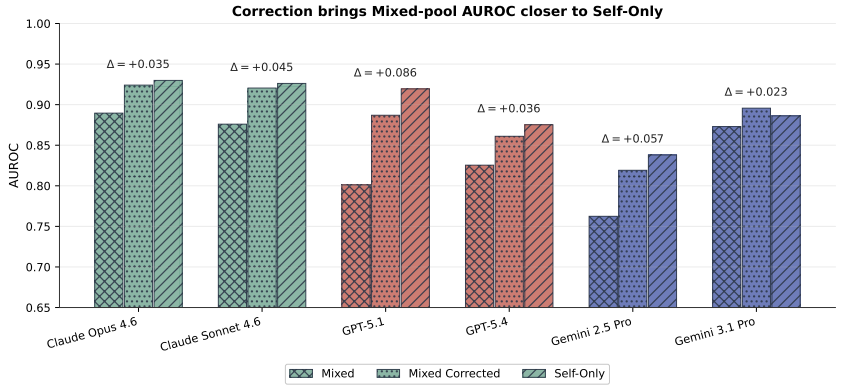


Figure 5: **Validation of the stratified harmonisation correction.** For each judge, AUROC under three conditions: **Mixed** (eval + all deploys except those generated by the judge model itself, raw scores), **Mixed Corrected** (same pool with harmonisation at $\lambda = 0.1$), and **self-only** (eval + same-family deploys, the unconfounded reference). Δ above each group is the AUROC lift from Mixed to Mixed Corrected. The corrected AUROC is consistently closer to self-only than the uncorrected version.

The blend weight is chosen conservatively because the harmonisation relies on per-stratum (mean, std) estimates, and some strata are small. At $\lambda = 0.1$ the harmonised value contributes at most

10% of any corrected score, bounding the influence of the smaller strata's noisy estimates; higher λ would push the corrected ranking closer to self-only but lets small-stratum noise propagate more aggressively into the corrected AUROC. Appendix E discusses this in more detail.

We validate this procedure in Figure 5. Across 6 models, we see that applying the correction brings results closer to the self-only measurement (which is the ideal but requires enough deployment data).

4 Evaluation Awareness Across Benchmarks and Models

We run the full pipeline: per-model PQ selection (Section 3.2) and stratified harmonisation (Section 3.3.3) on 12 judges in our suite, with the six closed-source frontier judges (Sonnet 4.6, Opus 4.6, GPT-5.1, GPT-5.4, Gemini 2.5 Pro, Gemini 3.1 Pro Preview) as the focus group where same-family deployment data is available for the harmonisation correction (See Section 3.3.3). We report: *AUROC*, an aggregate rank-based discrimination metric, and $P(\textit{eval})$, the raw per-transcript probability the judge assigns to the eval class.

Frontier models discriminate evaluation from deployment well above chance. Across all 12 judges (Figure 6), mixed AUROC ranges from 0.76 (Gemini 2.5 Pro, Claude Haiku 4.5) to 0.89 (Claude Opus 4.6), with every closed-source family showing improvement from older to newer release. The progression is most pronounced for the Anthropic family ($+0.13$ from haiku to opus).

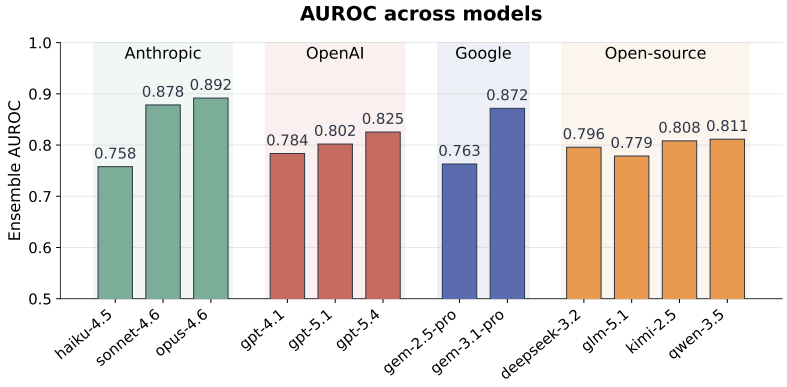


Figure 6: **Per-model mixed AUROC across 12 judges,** closed source models are grouped by family and ordered by release date. Within each closed-source family the newest model attains strictly the highest AUROC (Anthropic $0.76 \rightarrow 0.88 \rightarrow 0.89$; OpenAI $0.78 \rightarrow 0.80 \rightarrow 0.83$; Google $0.76 \rightarrow 0.87$). The four open-source judges cluster in $[0.78, 0.81]$. AUROC values here are uncorrected (no stratified harmonisation); see Figure 12.

$P(\textit{eval})$ is high across the benchmarks we measure. Pooled across the three frontier judges (GPT-5.4, Claude Opus 4.6 and Gemini 3.1 Pro Preview), mean $P(\text{eval}) > 0.83$ on a subset of benchmarks in our suite.

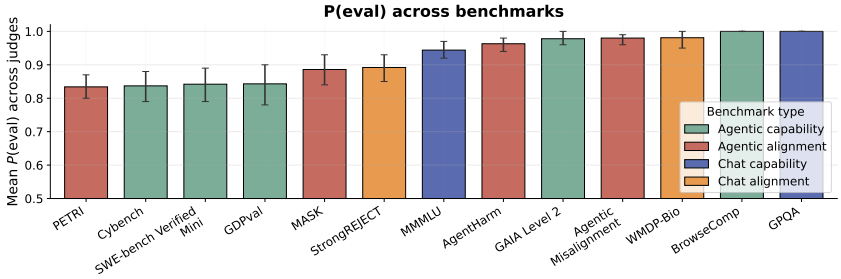


Figure 7: Mean $P(\text{eval})$ per evaluation dataset, averaged across six judges and grouped by category. Error bars are bootstrap 95% confidence intervals.

Selected probes surface different signals. GPT-5.4's motivation probe scores 0.00–0.05 on SWEBENCH-MINI and GDPVAL, while its origin probe scores > 0.89 on the same transcripts. Opus-4.6's `h_provenance_a` drops to 0.60 on MMMLU and CYBENCH, while its authenticity probe stays at 0.95–1.00 on the same transcripts. Gemini-3.1's lowest probe score is 0.90 (motivation on STRONGREJECT). Ensemble AUROC averages over these heterogeneous signals; per-probe and per-judge variance reveals structural disagreement that the aggregate conceals. Full per-(judge, probe, benchmark) values are in Table A.1.

Judges can disagree on transcripts where task realism and meta-evaluation signals point in different directions (Section A.1). On GDPVAL, this split recurs across the majority of transcripts, with GPT-5.4 more often reading through meta-evaluation framing to the underlying task structure, while Opus and Gemini-3.1 more often treat explicit evaluation signals as decisive.

5 Discussion

What prompted classification measures. Asking a model directly whether a transcript comes from an evaluation or deployment provides what we treat as a maximal-elicitation upper bound on evaluation awareness: if a model cannot distinguish the two even when explicitly asked, it is unlikely to do so spontaneously. This framing rests on two assumptions. The first, that asking does not itself inflate scores (the priming effect), we partially defend against through our choice of metric. The second, that the model answers honestly rather than dishonestly (i.e. sandbags [van der Weij et al., 2025]), remains untested in this work and in prior work, and is an open methodological problem for the field.

Priming effect. We chose AUROC as our primary metric specifically because it defends against the priming effect: a probe that explicitly mentions evaluation may inflate the likelihood of classifying a transcript as an eval, but as long as the inflation is roughly uniform across both classes, AUROC (which depends only on the rank ordering between eval and deploy transcripts) is unchanged. This is the central reason we report AUROC rather than raw $P(\text{eval})$ or accuracy at a fixed threshold. The defence is partial, however: priming is not necessarily uniform, and if it interacts with transcript content (e.g., inflating scores more on ambiguous deploy transcripts than on clear eval ones), the differential bias survives AUROC. Quantifying this residual interaction would require a probe condition that does not itself prime, which prompted classification cannot provide. This is the fundamental limitation of all prompted approaches, and the strongest argument for complementing them with white-box methods (Section 2) that do not require asking the model anything. Unfortunately this is not possible on closed-source frontier models.

Practical implications. Eval-awareness measurement should be standard practice alongside safety evaluations. A benchmark that is easily recognisable as an evaluation produces results whose validity is conditional: any safety case built on it must account for the possibility that models behaved differently than they would in deployment. Reporting eval-awareness scores alongside benchmark results strengthens the evidential chain in safety cases, and EvalDetectBench is designed to slot into existing Inspect-based workflows with minimal adoption friction.

Limitations. Beyond the priming and honest-reporting limitations discussed above, there are another two methodological limitations. First, our deployment pool is assembled from public sources and is unlikely to match any single provider's true traffic distribution. AUROC on this pool is a proxy for discrimination against real deployment data. Second, the stratified harmonisation (Section 3.3.3) is validated by agreement with the self-only condition on a disjoint deploy pool, but self-only is itself an imperfect proxy, and some strata might be small or noisy. We present harmonisation as a pragmatic correction rather than a debiased estimator with known error bounds.

Conclusion. We showed that standard methodological choices in evaluation awareness measurement, including transcript sourcing, probe question selection, and scoring format, introduce biases that, if uncontrolled, can be larger than the differences in eval-awareness between the models being measured. Generator identity contributes 11.25% of $P(\text{eval})$ variance in heterogeneous deployment pools, and probe-question sets calibrated on one model family under-elicit awareness in others by up to 0.08 AUROC. EvalDetectBench incorporates these findings into an open pipeline that supports per-model probe calibration and generator-confound correction for any Inspect-compatible evaluation.